

An AI-ML model for air quality prediction of Indian cities

1. Project description and objectives

Build an AI-ML model for air quality prediction of Indian cities: The objective of this project is to quantitatively analyze the relationship between air quality indicators (AQI, PM_{2.5}, and PM₁₀) and key meteorological drivers across major Indian cities using available datasets from 2019 to the present, and to develop an AI-ML framework for reliable air-quality prediction. The proposed workflow includes: integrating multi-year air pollution and meteorological datasets to construct a harmonized, quality-controlled database for robust temporal and spatial analysis; quantifying the magnitude and uncertainty of meteorological influences on pollution dynamics and identifying weather conditions associated with high-pollution episodes; developing AI/ML model(s) that combine meteorological variables with historical air-quality data to predict future pollution levels; training, validating, and testing the model(s) using appropriate data-partitioning and cross-validation strategies, with performance evaluated on independent test datasets using standard statistical metrics; generating short-term and long-term air-quality forecasts using the trained models; and assessing model robustness through spatial evaluation across monitoring locations within cities and testing generalization across multiple cities. The outlined workflow is indicative, and candidates are encouraged to adopt alternative or improved methodologies as appropriate. Use your imagination!

Relevant literature/Starting work

Refer to the attached publications for getting started. The expectations are to use these works as a starting point and build models which can go beyond what is already published.

Team formation and communication

Class is divided into groups, and each group is assigned with an Indian city, which you will use for your analysis. The group consists of up to 5 members and a project communicator. The communicator is responsible for communicating, i.e., uploading the final project report and final presentation.

Data sources

AQI, pollutant, meteorological data: Please download the daily data from the CPCB NAMP / AQI Bulletin Archive (either station-level or city-level aggregates) from 2019 to date. Retrieve as much available data as possible for the relevant/chosen city/location.

Primary source - CPCB. [<https://airquality.cpcb.gov.in/ccr/#/caaqm-dashboard-all/caaqm-landing/caaqm-data-repository>]

Optional Data (Advanced / Cross-check):

Meteorology:

CPCB website also provides meteorological data (temperature, humidity, wind, rainfall,...).

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You may also explore **NASA POWER** datasets for the same period (2019–2025) to cross-check values or fill in missing data. [<https://power.larc.nasa.gov/data-access-viewer/>], but rely on CPCB first.

Tools & environment

Google Colab (free, GPU not needed + easy sharing).

Python stack: **pandas**, **numpy**, **matplotlib/plotly**, **seaborn** (optional), **scikit-learn**, **statsmodels**.

Version control: share notebooks via GitHub or Google Drive.

Workflow (suggested notebooks)

Data download & preprocessing

Download CPCB/AQI data by station/year (2019 to date). Aggregate to daily or hourly means as required.

Pull NASA POWER daily data for each city coordinate and date range.

Align by date and location; handle missing values (impute or flag); standardize units.

Exploratory Data Analysis (EDA)

Time-series plots: daily/monthly AQI and pollutant trends (PM_{2.5}, PM₁₀, NO₂, SO₂, O₃, CO).

Seasonal decomposition; identify peaks (festival/fireworks, crop burning, lockdown 2020 anomalies).

Spatial comparison across cities (boxplots, heatmaps).

Statistical & ML analysis

Correlations (Spearman + Pearson) between pollutants/AQI and meteorological variables.

Simple regression: pollutant $\sim$ wind + temp + RH + rainfall.

ANN for prediction.

Interpretation & sensitivity

Quantify meteorological influence (e.g., % variance explained).

Discuss confounders (emissions changes, lockdowns, local sources).

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Visualizations & presentation

Required visuals: time-series per city, seasonal boxplots, correlation matrix, map (city markers), and model diagnostics.

Presentation: **7 minutes of pre-recorded video presentation + 3 minutes live Q&A**. Each slide deck: 12 slides maximum.

Deliverables (per group)

Cleaned merged dataset (CSV / notebook): AQI + meteorology.

EDA visuals: PNG/embedded in notebook.

Analytical results: correlation tables, regression outputs, feature importances.

Short report (2–4 pages): Methods, key findings, limitations.

Slide deck: 12 slides maximum

Presentation: 7-minute pre-recorded audio/video presentation preferably < 200 MB

Outline of grading scheme (out of 100)

Data quality & preprocessing: 20

EDA & visual clarity: 20

Analysis rigor (models/statistics): 25

Interpretation & discussion: 20

Presentation & teamwork: 15



Development of ANN model for Prediction of AQI at Sanathnagar, Hyderabad, India.

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KEYWORDS

Artificial Neural
Networking, Air
Quality Index,
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ABSTRACT:

Air pollution is not only environmental problem, but also a major health issues in many countries. The effective methods in controlling the air pollution is by a proper air quality management. One of effective methods is by predicting the air quality index. One such method is based on prediction of data is by using Artificial neural networking (ANN). In this study 21 models were developed for prediction of pollutants. The required data for development of ANN models were collected from the TSPCB (Telangana State Pollution control board), Hyderabad, India. The data comprises 2007-2017 years of hourly data. Best suited models were selected on the basis of statistical tools such as R and MSE. This study was conducted seasonally-Summer, Post-monsoon, Winter. The predicted data is for 2017 – 2023. The AQI (predicted) is Computed by using IND-AQI CPCB method and the quality rating is giving by color coding. After the prediction a comparison was drawn between AQI (observed) and AQI (predicted), then the conclusions were drawn from this table.

1. Introduction

Air quality management is a crucial process that involves monitoring air quality, assessing the impact of human activities, taking measures to improve the situation, and ensuring that these measures are effective. This system is designed to reduce the emission of pollutants and other harmful substances in the atmosphere and to sustain ambient air quality. In recent years, effective methods have been developed globally to understand and summaries the importance of good air quality. One of the best ways to maintain air quality is by computing the air quality index, which helps to categorize the meteorological conditions relevant to the Air Quality of that country or city.

In recent times many methods have been developed for prediction of AQI. Recent Studies have proving that ANN method is the most interesting approach to predict AQI.

ANNs are used in complex situations, especially in predicting air pollutants, to provide better results. ANN models can predict air quality over various time periods, making them ideal for predicting regional or global air quality during specific periods. ANNs differ from traditional methods in terms of mapping variables, prediction accuracy, and handling missing values. ANNs can be categorized into two types: Feed-forward networks and recurrent networks. Feed-



forward networks have one layer of hidden neurons, while recurrent networks have at least two layers of hidden neurons.

When describing air quality, it is important to report the concentration of all pollutants in that location. Air quality parameters include gases, volatile organic compounds, and meteorological parameters. In India, rapid urban development has led to severe crises in air quality, affecting biology, physics, and economic system. Based on the above passages, developing an approach to interpret and predict future air concentrations in urban environment is urgently required. One such work is aimed at in this study.

2. Objective of the Study

The primary objective of this study is to predict the ambient air quality of the study area by developing an artificial neural network model using real time data and finding the significant meteorological factors affecting the air pollutants concentration in a given period.

3. Study area

The study area Sanathnagar is both industrial cum residential where the industrial area is home to a variety of small and medium scale chemical, pharmaceutical, electrical and mechanical industries. The connectivity of study area to neighboring places is via road, which implies that most of the pollution concentrations are related to vehicular emissions. These conditions make it an ideal study area for understanding and predicting ambient air quality. The Telangana state Pollution Control Board (TSPCB) provides the hourly atmospheric concentration of all the pollutants that fall under the Central Pollution Control Board (CPCB).

4. Data collection and aggregation

The data relating to the sampling location Sanathnagar is collected from TSPCB, Hyderabad, Telangana, India for the years 2007-2017. The following Table 1 describes the types of parameters collected from TSPCB for this study.

Table 1: Type of parameters collected

S.No	Parameters	Type of parameters	Unit
1	Carbon Monoxide CO	Ambient air pollutant	$\frac{\mu g}{m^3}$
2	Sulphur dioxide SO ₂	Ambient air pollutant	$\frac{\mu g}{m^3}$
3	Particulate Matter PM	Ambient air pollutant	$\frac{\mu g}{m^3}$
4	Oxides of Nitrogen NO _x	Ambient air pollutant	$\frac{\mu g}{m^3}$
5	Benzene C ₆ H ₆	Ambient air pollutant	$\frac{\mu g}{m^3}$
6	Toluene C ₆ H ₅ CH ₃	Ambient air pollutant	$\frac{\mu g}{m^3}$
7	Xylene (CH ₃) ₂ C ₆ H ₄	Ambient air pollutant	$\frac{\mu g}{m^3}$
8	Atmospheric Temperature AT	Meteorological Parameter	°C
9	Relative Humidity RH	Meteorological Parameter	%
10	Wind Speed WS	Meteorological Parameter	$\frac{m}{s}$



11	Sun radiation SR	Meteorological Parameter	$\frac{W}{m^2}$
12	Wind Direction WD	Meteorological Parameter	deg

5. Artificial Neural network (ANN) modelling

ANN modelling has several advantages over statistical methods with an accurate prediction. The objective of this study is to implement ANN modelling for predicting air quality based on the previously available data. Therefore, in this modelling the prediction is carried out for the air pollutants such as CO, SO₂, NO_x, PM, Benzene, Toluene, and Xylene. The prediction is carried out by using meteorological parameters as set of inputs. here the input data is obtained

from Principal component analysis and Cluster analysis outputs.

The air quality data obtained from TSPCB is divided into two parts: Seasonal data from 2007 to 2016, used for model development and training of the network, whereas 2017 data, used for additional testing of the network. Therefore, the model prediction is given for 2017 to 2026, and the predicted data was validated by comparing with 2017 (observed) data.

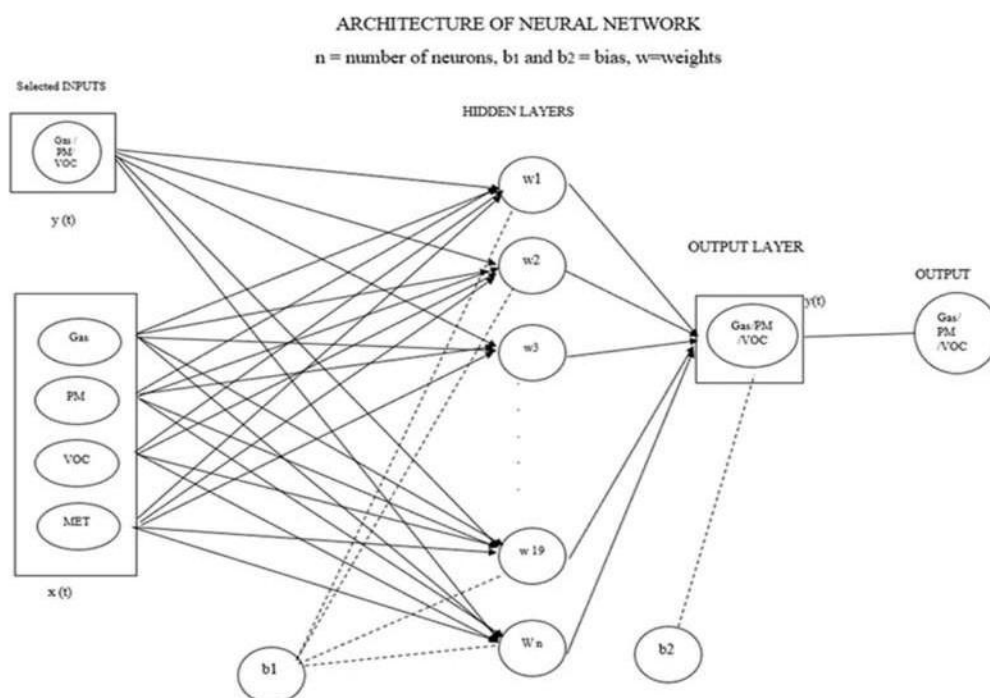


Figure 1: Arcitecture of Nueral Network



6. Results and Discussion

6.1 Architecture of the neural network

This network model utilizes non-linear time series prediction and modelling tool (ntstool) with two layered architecture.

The network is Non-Linear auto regression exogenous model (NARX) network, this network predicts the series $y(t)$ based on the past values of $y(t)$ and another series $x(t)$. This is feed-forward network.

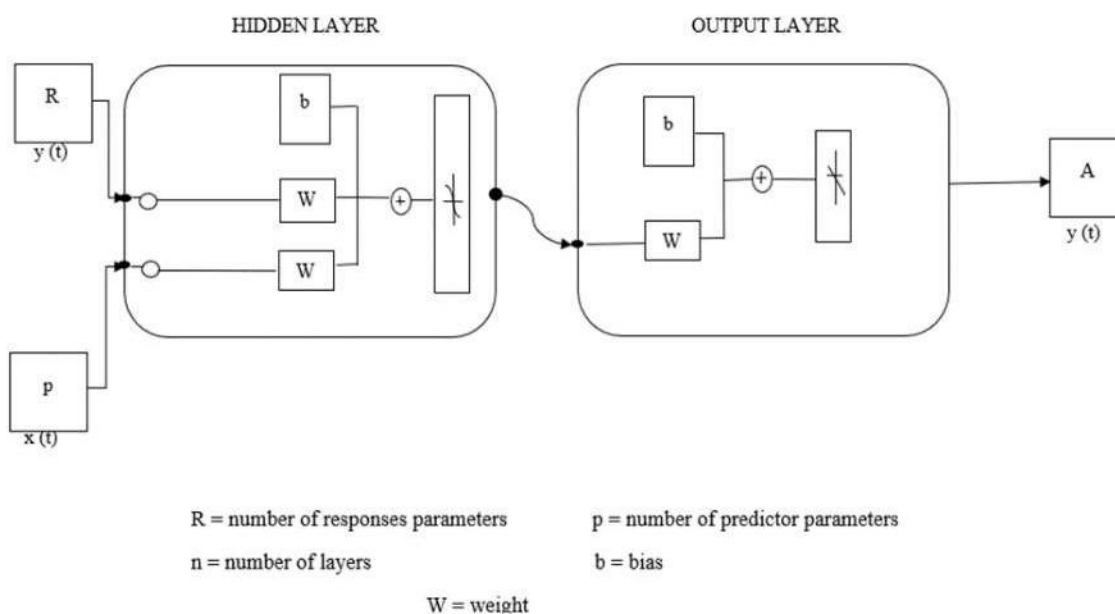


Figure 2: NARX Network Model

The figure 2 displayed above represents the model structure developed in this research shows a two layered simple feed forward network. It comprises of four sections. The first section includes $y(t)$ and $x(t)$. The second section consists of a hidden layer that comprises of weights, bias neurons, and input variables. This is where the network training takes place. The third section includes output variables that are influenced by the weights and bias. This is where the network validation and testing are conducted. The final stage involves obtaining predicted variables through prediction.

The input data for modelling varies based on the season and model. The number of neurons and hidden layers in the network architecture effect the output, therefore different sets of neurons (n) are used, specifically 10,15 and

20. Here the tool has segregated the input data for each model is segregated into three categories training data (70%), validation data (15%), and test data (15%), with a time delay of 2. The training algorithm: Levenberg- Marquardt is applied for all the models.

6.2 Selected ANN models for Data Prediction

A total of 180 ANN models were created for all the three seasons to predict and develop AQI in the study area. Out of 180 models, 21 were selected because they had the best suited values of R and MSE. The selection was based on accuracy, efficiency, and the ability to handle large data sets, and by taking into consideration the statistical values of both MSE, R and testing of the network. The Table 2 consist of the structural architecture of the different parameters as follows



“IN-n-OP”, where IN = number of input variables, n= number of layers, OP= number of output variables. The time

mentioned in the table head indicates the time required by the model to produce results, which is in minutes.

Table 2: ANN models for Data Prediction

S.NO	SEASON	PARAMETER	NETWORK ARCHITECTURE (IN-n-OP)	TIME (MIN)	TRAINING RESULTS	TESTING RESULTS
1	SUMMER	CO	4-20-1	0	MSE = 0.034 R=0.749	MSE =0.034 R=0.673
		SO ₂	4-15-1	0	MSE =34.323 R=0.7161	MSE = 4.616 R=0.3302
		NO _x	6-10-1	0	MSE = 455.33 R=0.7852	MSE =260.76 R=0.7376
		PM	6-20-1	0	MSE =842.9 R=0.854	MSE =355.79 R=0.741
		BENZENE	4-15-1	0	MSE =1.152 R= 0.858	MSE = 0.235 R= 0.230
		TOLUENE	4-20-1	0	MSE =130.33 R= 0.818	MSE = 55.186 R= 0.3011
		XYELENE	4-20-1	0	MSE = 3.028 R= 0.881	MSE = 0.9768 R= 0.290
2	POST-MONSOON	CO	5-15-1	0	MSE = 0.027 R=0.698	MSE = 0.029 R=0.029
		SO ₂	5-15-1	0	MSE =61.68 R=0.859	MSE = 15.327 R=0.0095
		NO _x	7-15-1	0	MSE = 516.37 R=0.775	MSE = 266.90 R=0.270
		PM	4-15-1	0	MSE = 861.35 R=0.900	MSE =303.83 R=0.743
		BENZENE	6-15-1	0	MSE =18.722 R=0.957	MSE = 2.089 R=0.150
		TOLUENE	6-15-1	0	MSE =306.69 R=0.983	MSE = 183.50 R=0.457
3	WINTER	CO	4-15-1	0	MSE =0.0269 R=0.814	MSE =0.042 R=0.620
		SO ₂	4-15-1	0	MSE = 87.27 R=0.795	MSE = 80.780 R=0.780
		NO _x	6-20-1	0	MSE =674.53 R=0.846	MSE =602.13 R=0.611
		PM	4-10-1	0	MSE	MSE = 509.94



					=1.0170e+03 R=0.885	R=0.555
BENZENE	6-20-1	0	MSE = 13.929 R=0.854	MSE =2.331 R=0.7910		
TOLUENE	4-20-1	0	MSE = 389.54 R=0.793	MSE = 190.618 R=0.795		
XYELENE	6-20-1	0	MSE =7.436 R=0.916	MSE =2.60 R=0.628		

The three plots used for understanding the accuracy of the model, are the performance plot, error histogram plot and the response plot.

6.3 Comparison study on AQI Observed and Predicted

The comparison is observed in the tabular form to facilitate seasonal study. Therefore, two set of data sets are used for the same purpose where the predicted data is observed from 2024

– 2026 and for the verification of model is done with the data 2017-2023. For the verification purpose a set of AQI data (observed) was downloaded from the CPCB website.

The comparison of AQI is shown in the below mentioned table 3, this table consist of observed and predicted values of AQI for all the three seasons.

Table 3: AQI: Predicted and Observed Value

Season	Year	Month	AQI Predicted		AQI CPCB	
			Value	Quality status	Value	Quality status
SUMMER	2017	MARCH	128	MODERATE	90	SATISFACTORY
		APRIL	91	SATISFACTORY	107	MODERATE
		MAY	79	SATISFACTORY	73	SATISFACTORY
		JUNE	62	SATISFACTORY	38	GOOD
	2018	MARCH	67	SATISFACTORY	122	MODERATE
		APRIL	99	SATISFACTORY	83	SATISFACTORY
		MAY	102	MODERATE	69	SATISFACTORY
		JUNE	99	SATISFACTORY	45	GOOD
	2019	MARCH	117	MODERATE	89	SATISFACTORY
		APRIL	110	MODERATE	69	SATISFACTORY
		MAY	107	MODERATE	77	SATISFACTORY
		JUNE	57	SATISFACTORY	52	SATISFACTORY
	2020	MARCH	111	MODERATE	66	SATISFACTORY
		APRIL	92	SATISFACTORY	56	SATISFACTORY
		MAY	75	SATISFACTORY	63	SATISFACTORY
		JUNE	57	SATISFACTORY	34	GOOD
	2021	MARCH	143	MODERATE	128	MODERATE
		APRIL	117	MODERATE	90	MODERATE
		MAY	140	MODERATE	48	GOOD



Season	Year	Month	AQI Predicted		AQI CPCB	
			Value	Quality status	Value	Quality status
	2022	JUNE	88	SATISFACTORY	36	GOOD
		MARCH	62	SATISFACTORY	123	MODERATE
		APRIL	116	MODERATE	81	SATISFACTORY
		MAY	120	MODERATE	74	SATISFACTORY
		JUNE	63	SATISFACTORY	47	GOOD
	2023	MARCH	143	MODERATE	74	SATISFACTORY
		APRIL	117	MODERATE	82	SATISFACTORY
		MAY	140	MODERATE	67	SATISFACTORY
		JUNE	88	SATISFACTORY	46	GOOD
POST-MONSOON	2017	OCTOBER	114	MODERATE	113	MODERATE
		NOVEMBER	135	MODERATE	107	MODERATE
	2018	OCTOBER	123	MODERATE	86	SATISFACTORY
		NOVEMBER	145	MODERATE	163	MODERATE
	2019	OCTOBER	116	MODERATE	93	SATISFACTORY
		NOVEMBER	108	MODERATE	188	MODERATE
	2020	OCTOBER	79	SATISFACTORY	103	MODERATE
		NOVEMBER	107	MODERATE	126	MODERATE
	2021	OCTOBER	116	MODERATE	94	SATISFACTORY
		NOVEMBER	134	MODERATE	106	MODERATE
	2022	OCTOBER	102	MODERATE	97	SATISFACTORY
		NOVEMBER	121	MODERATE	178	MODERATE
WINTER	2017	JANUARY	125	MODERATE	173	MODERATE
		FEBRUARY	114	MODERATE	203	POOR
		DECEMBER	129	MODERATE	243	POOR
	2018	JANUARY	125	MODERATE	229	POOR
		FEBRUARY	110	MODERATE	125	MODERATE
		DECEMBER	192	MODERATE	202	POOR
	2020	JANUARY	146	MODERATE	211	POOR
		FEBRUARY	140	MODERATE	100	MODERATE
		DECEMBER	122	MODERATE	176	MODERATE
	2021	JANUARY	107	MODERATE	104	MODERATE
		FEBRUARY	103	MODERATE	95	SATISFACTORY
		DECEMBER	123	MODERATE	192	MODERATE
	2022	JANUARY	122	MODERATE	159	MODERATE
		FEBRUARY	136	MODERATE	158	MODERATE
		DECEMBER	136	MODERATE	185	MODERATE
	2023	JANUARY	138	MODERATE	141	MODERATE
		FEBRUARY	147	MODERATE	107	MODERATE



Season	Year	Month	AQI Predicted		AQI CPCB	
			Value	Quality status	Value	Quality status
		DECEMBER	110	MODERATE	155	MODERATE

The Central Pollution Control Board (CPCB) has defined an AQI of 50 as fairly clean atmospheric conditions. However, it has been observed that the predicted AQI for all three seasons has been consistently above this value throughout the study period. This indicates that the air quality in the region needs improvement.

7. Conclusion

Out of the three seasons, winter can be named as the worst season or highly polluted season with the worst air quality rating. After the Table 3 observation and the quality rating table it can be concluding that the model is "correct" and giving satisfactory results as both the predicted and observed status are similar.

8. Acknowledgement

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Air pollution prediction by using an artificial neural network model

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Abstract

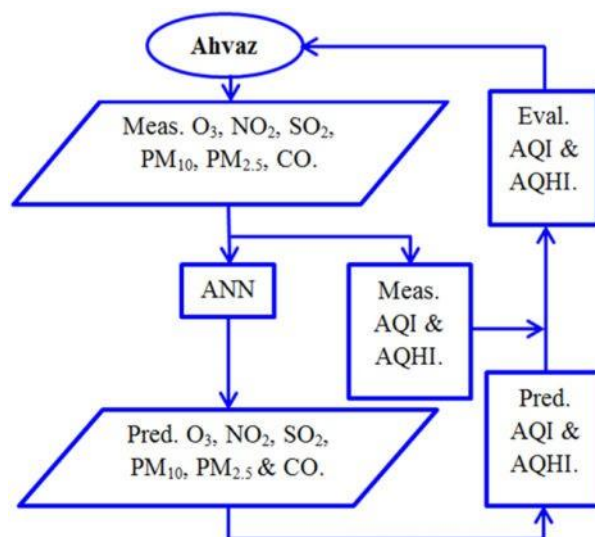
Air pollutants impact public health, socioeconomics, politics, agriculture, and the environment. The objective of this study was to evaluate the ability of an artificial neural network (ANN) algorithm to predict hourly criteria air pollutant concentrations and two air quality indices, air quality index (AQI) and air quality health index (AQHI), for Ahvaz, Iran, over one full year (August 2009–August 2010). Ahvaz is known to be one of the most polluted cities in the world, mainly owing to dust storms. The applied algorithm involved nine factors in the input stage (five meteorological parameters, pollutant concentrations 3 and 6 h in advance, time, and date), 30 neurons in the hidden phase, and finally one output in last level. When comparing performance between using 5% and 10% of data for validation and testing, the more reliable results were from using 5% of data for these two stages. For all six criteria pollutants examined (O₃, NO₂, PM₁₀, PM_{2.5}, SO₂, and CO) across four sites, the correlation coefficient (R) and root-mean square error (RMSE) values when comparing predictions and measurements were 0.87 and 59.9, respectively. When comparing modeled and measured AQI and AQHI, R^2 was significant for three sites through AQHI, while AQI was significant only at one site. This study demonstrates that ANN has applicability to cities such as Ahvaz to forecast air quality with the purpose of preventing health

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effects. We conclude that authorities of urban air quality, practitioners, and decision makers can apply ANN to estimate spatial–temporal profile of pollutants and air quality indices. Further research is recommended to compare the efficiency and potency of ANN with numerical, computational, and statistical models to enable managers to select an appropriate toolkit for better decision making in field of urban air quality.

Graphical abstract



Keywords

Criteria air pollutants; ANN; AQI; AQHI

Introduction

Air pollution poses deleterious effects on people's health, especially those in vulnerable populations such as children, elder women and men, and patients who suffer from respiratory and cardiovascular diseases (Naddafi et al. 2012; Nourmoradi et al. 2016). Air pollution also has detrimental effects on the environment, socioeconomics, agriculture, and politics (Zhang et al. 2008; Vlachokostas et al. 2010; Maghrabi et al. 2011; Hou et al. 2016). The following six criteria air pollutants are sufficiently harmful for humans and the environment that they are routinely monitored by the United States Environmental Protection Agency (US EPA), which has set National Ambient Air Quality Standards (40 CFR part 50) for these species: carbon monoxide, lead, nitrogen dioxide, ozone, particulate matter (PM₁₀ and PM_{2.5}), and sulfur dioxide. The EPA reports daily air quality conditions as an air quality index (AQI), which is calculated based on these criteria air pollutants (except lead). The air quality health index (AQHI), developed by Health Canada and Environment Canada (HCEC), is an analogous index considering PM_{2.5}, O₃, and NO₂.

Meteorology and emissions sources of pollutants are two basic factors influencing the aforementioned air quality indices and can be used in computational approaches to predict

spatiotemporal pollutant profiles and air quality index values. Various applicable examples include the Community Multi-scale Air Quality (CMAQ) model, the Weather Research and Forecasting model with Chemistry (WRF-Chem), and artificial neural network (ANN) and fuzzy inference systems (Pokrovsky et al. 2002; Cai et al. 2009; Wang et al. 2010; Feng et al. 2015; Zhang et al. 2015, 2016). ANN approaches can enhance forecasting accuracy relative to previously used statistical procedures (Nagendra and Khare 2005). Types of ANN include the back-propagation neural network (BPNN) (Chen and Pai, 2015; Bai et al. 2016), multilayer perceptron (MLP) (Wang and Lu 2006; Durão et al. 2016), radial basis function (RBF) (Lu et al. 2004; Iliyas et al. 2013), and adopted neuro-fuzzy inference systems (ANFIS) (Shahraiyini et al. 2015; Prasad et al. 2016). Among the earliest applications of ANN for air pollution, research was forecasting SO₂ levels in Slovenia (Boznar et al. 1993). More recent attempts of using ANN include combining remotely sensed aerosol optical depth (AOD) and meteorological data to estimate surface PM_{2.5} levels (Gupta and Christopher 2009). Numerous studies have evaluated various aspects of PM_{2.5} and PM₁₀ mass concentration in different areas such as Taiwan (Chang and Lee 2007), Italy (Ragosta and Gioscio 2009), New Zealand (Elangasinghe et al. 2014), the western Mediterranean (de Gennaro et al. 2013), India (Patra et al. 2016), Portugal (Russo et al. 2015), and China (Qin et al. 2014; Yao and Lu 2014).

In this study, we apply a ANN model approach to predict hourly criteria air pollutant concentrations (O₃, NO₂, SO₂, PM₁₀, PM_{2.5}, CO), daily AQI, and hourly AQHI. The analysis focuses on Ahvaz, Iran, which suffers from some of the worst air conditions globally owing to severe dust storms. It was recently reported to be the highest ranked city in terms of mean-annual PM₁₀ concentration on the Earth (372 µg m⁻³; Goudie 2014). The subsequent analysis focuses on model prediction comparisons to measurement data at four sites around Ahvaz, with attention given to how well diurnal profiles are reproduced.

Experimental methods

Study area

Ahvaz is the capital of Khuzestan province located close to the Persian Gulf in southwestern of Iran, and it contains the largest river of Iran (Karun) (Fig. 1). Its population is ~ 1.2 million and has an area of ~ 528 km² (Naimabadi et al. 2016). The climate in this region is hot and humid. This study makes use of hourly data collected between August 2009 and August 2010 by the Ahvaz Environmental Protection Organization (AEPO) and Ahvaz Meteorological Center. Samples were obtained using the beta attenuation procedure, which is often common at routine monitoring stations. Measurements of O₃, NO₂, CO, SO₂, and PMs were conducted, respectively, via the use of Beer–Lambert’s law, chemiluminescence, non-dispersive infrared spectroscopy, UV fluorescence, and beta radiation (Mazaheri Tehrani et al. 2015; Alizadeh-Choozari et al. 2016). Data related to pollutants concentration and meteorological parameters affecting the forecast of model, such as wind speed (m/s), ambient air temperature (°C), dew point (°C), rainfall (mm), and air pressure, are presented in Table 2. The four sampling locations for O₃, NO₂, PM₁₀, PM_{2.5}, SO₂, CO, and meteorology are shown in Fig. 1 (Naderi, Havashenasi, Mohite Zist, Behdasht).

Air pollution indexes

AQI is calculated using concentration data for PM, NO₂, SO₂, CO, and O₃. Index values differ among global cities owing to various levels of anthropogenic activity, potential sources of natural emissions, as well as transport patterns of pollutants. Values are classified as good (0–50, green), moderate (51–100, yellow), unhealthy for susceptible groups (101–150, orange), unhealthy (151–200, red), very unhealthy (201–300, purple), and hazardous (301–500, Maroon). As will be shown, AQI values exceed 500 in the study region. The general equation to calculate AQI is as follows:

$$I_p = \frac{I_{Hi} - I_{Lo}}{BP_{Hi} - BP_{Lo}} (C_p - BP_{Lo}) + I_{Lo} \quad (1)$$

where I_p = the index for pollutant p , C_p = the rounded concentration of pollutant p , BP_{Hi} = the breakpoint that is $\geq C_p$, BP_{Lo} = the breakpoint that is $\leq C_p$, I_{Hi} = the AQI value corresponding to BP_{Hi} , and I_{Lo} = the AQI value corresponding to BP_{Lo} .

AQHI presents the health risk of pollutants based on the following scale: low risk (1–3, blue), moderate risk (4–6, orange), high risk (7–10, red), and very high risk (> 10 , black). The AQHI is calculated according to the following formula using 3-h average concentrations of O₃, NO₂, and PM_{2.5} (El-Latef et al. 2018).

$$AQHI = \frac{1000}{10.4} * \left(e^{0.000537 * O_3} - 1 + e^{0.000871 * NO_2} \left(\left(1 - e^{0.000487 * PM_{2.5}} \right) \right) \right) \quad (2)$$

Artificial neural network

A neural network utilizes artificial neurons, which are the smallest units of data processing (Sadorsky 2006). The framework of a one-input system is demonstrated in Fig. 2. Equation 3 is used to quantify the output:

$$p = f \left(\sum_{i=1}^m w_i x_i \right) + b \quad (3)$$

$$\sigma(p) = \frac{1}{1 + e^{-p}} \quad (4)$$

where b and w represent parameters that are set based on a selected activation function and type of learning algorithm. A number of different activation functions such as linear, sigmoid, and hyperbolic tangent can be used that for this study Sigmoid function was used (Eq. 4) (Alimissis et al. 2018; Elfving et al. 2018).

Figure 3 depicts the ANN structure applied in this study. Nine input factors are used, with 30 nerve cells in the intermediate step leading to one outcome. The 30 neurons in the interior layer were identified through a trial-and-error process using between 6 and 60 nerve cells. It was necessary to determine what percentage of randomly selected data points from the

available sample set needed to be assigned to training, validation, and testing in order to obtain the best agreement between predicted and measured concentrations. Points assigned to training are used for computations and updating the network weights and biases to train candidate algorithms. For evaluation of performance, Eqs. 5 and 6 are used to obtain linear correlation coefficients and RMSE, respectively:

$$R^2 = \frac{\left[\sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}}) (Y_i - \bar{Y}) \right]^2}{\left[\sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2 \right]^{0.5}} \quad (5)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (Meas - Pred)^2}{n}} \quad (6)$$

where Y_i and $\bar{Y}$ are the measured concentrations and average of measured concentrations for a pollutant, respectively. $\hat{Y}_i$ and $\bar{\hat{Y}}$ are predicted concentrations and average of predicted concentrations for a pollutant, respectively, n is the number of data values, $Meas$ is the measured data, and $Pred$ is the predicted data. The ideal value for R^2 and RMSE is 1 and 0, respectively (Ho et al. 2002).

Results and discussion

It was first necessary to determine under what conditions the best agreement could be attained between predicted and measured concentrations. The ANN was tested with having either 5% or 10% of data used for both validation and testing, with the remainder for training. When the total data used in the validation and testing stage decreased from 10 to 5% for the four air quality monitoring stations in Ahvaz, the average correlation coefficient

(R) between measured and predicted concentrations (for all pollutants and sites combined) increased by 8.1% while RMSE decreased by 11.8% (Table 1).

Figure 4 shows the relationship between measured and predicted concentrations at all sites for different pollutants based on 5% of data for validation and testing stages. Among criteria air pollutants, SO_2 exhibited the best results at the Mohite Zist station ($R = 0.99$), while SO_2 yielded the worst performance at the Havashenasi station ($R = 0.75$). The lowest RMSE was for CO at the Havashenasi station (0.3), while the highest value was for PM_{10} at the Mohite Zist station (360.5). The averages among all pollutants and sites combined for R and RMSE were 0.87 and 59.9, respectively.

A comparison between measured and predicted pollutant concentrations of all four stations is presented in Table 2. When taking an average for all six pollutants, the range of variation for predicted levels increased by 3.4% as compared to measured values, whereas the standard deviation decreased by 10.2%. No difference was observed between the average of total predicted and measured pollutant concentrations. The widest range of values was for PM_{10} , while the lowest was for CO.

The diurnal concentration profile of measured and predicted pollutant concentrations at the four air quality monitoring stations in Ahvaz is presented in Fig. 5. Expectedly based on the results of Table 1, the diurnal profiles for the studied pollutants are similar between measurements and the model. O₃, PM₁₀, PM_{2.5}, and SO₂ exhibit minimum levels early in the morning and late night, with maximum values in between during the day. Diurnal variations for NO₂ and CO exhibit minimum values in the early morning and afternoon, and maximum levels before noon and at night.

A comparison between calculated and predicted AQI is presented in Table 3 for the four sites in Fig. 1 and also for all four sites combined, representative of the entire city of Ahvaz. With regard to R^2 values, the ANN method was weak in terms of predicting AQI at the Naderi, Mohite Zist, and Behdasht stations, but it had acceptable performance ($R^2 = 0.73$) at the Havashenasi station. The cumulative percentage of days where the predicted AQI category did not match the measured one at Naderi, Havashenasi, Mohite Zist, and Behdasht stations and the bulk average of all sites as Ahvaz city was 49.6%, 38.1%, 61.3%, 48.5%, and 42.7%, respectively. Also, the number of days when at least one of the six pollutants exceeded its maximum limit at Naderi, Havashenasi, Mohite Zist, and Behdasht, and Ahvaz city was 28, 23, 31, 25, and 71, respectively.

The daily profile of AQI during the study period (August 2009–August 2010) is illustrated in Fig. 6. We assumed the upper limit of AQI was equal to 600 instead of 500. At the Havashenasi station, there were no significant fluctuations for AQI in November, December, and January. Averages of calculated AQI were 188, 178, 172, 180, and 240 at Naderi, Havashenasi, Mohite Zist, and Behdasht, and Ahvaz city, respectively, without considering (by ignoring) the assumed value of AQI for correspondent pollutants during the study period. As a result, the capability of ANN for predicting AQI increased when the number of out-of-range days related to correspondent pollutants (mostly PM₁₀ and PM_{2.5} in Ahvaz) decreased. By considering the maximum level of AQI at all stations to describe Ahvaz's AQI, the average of it in the city of Ahvaz was 1.34 times higher than mean AQI of air quality control stations.

A comparison between calculated and predicted AQHI is presented in Table 4. The R^2 value from the ANN method for predicting AQHI was 0.81, 0.70, and 0.78 at Naderi, Mohite Zist, and Behdasht, respectively, indicative of a decent level of performance. The Havashenasi station was an exception with a low R^2 value, which implies weak performance of the ANN to predict AQHI at this station. The percentage of hours with mismatches between predicted and measured AQHI at Naderi, Havashenasi, Mohite Zist, and Behdasht was 35.7%, 45.6%, 40.9%, and 37.8%, respectively. Also, the number of hours for AQHI being greater than or equal to 20, which is close to an AQI of 500, was 553, 677, 450, and 382 (out of a total of 8030) at Naderi, Havashenasi, Mohite Zist, and Behdasht, respectively.

The temporal profile of calculated and predicted AQHI at the four stations over a span of 1 year (8760 h) is depicted in Fig. 7. On average, the calculated AQHI values at Naderi, Havashenasi, Mohite Zist, and Behdasht were 10, 10, 8, and 9, respectively. If there are many hours with $AQHI \geq 20$, the capability of ANN to accurately predict AQHI decreases.

Of note is that there was no significant fluctuation in Ahvaz during December and January for AQHI.

Conclusions

The overall correlation coefficient (R) for measured versus predicted pollutant concentrations for several sites in Ahvaz was shown to be 0.87 over the span of 1 year. The diurnal concentration profiles for both measured and predicted O_3 , PM_{10} , $PM_{2.5}$, and SO_2 were similarly unimodal, while the variations for NO_2 and CO were bimodal. The critical aspect for the accuracy of ANN model is the number of high-polluted hours and days, respectively, for AQHI and AQI.

We conclude that authorities of urban air quality, practitioners, and decision makers can apply ANN to forecast the spatial-temporal profile of pollutant concentrations and air quality indices during power outage and wrong and negative records of pollutants. The study proposes that the ANN models could be used as an effective alternative in air pollution spatial interpolation and the representativeness of the air monitoring networks by providing data at currently unmonitored locations and thus eliminating the requirement of a relatively high number of monitoring stations for describing the air pollution spatial variability (Alimissis et al. 2018). Further research is recommended to compare the efficiency and potency of ANN with numerical, computational, and statistical models to enable managers to select an appropriate toolkit for better decision making in field of urban air quality.

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Fig. 1.

Map showing the location of four sites from which data were collected for criteria air pollutants in Ahvaz, Iran. Names of sites (from top to bottom) are Havashenasi (Meteorology), Naderi, Behdasht, and Mohit Zist (Maleki et al. 2016)

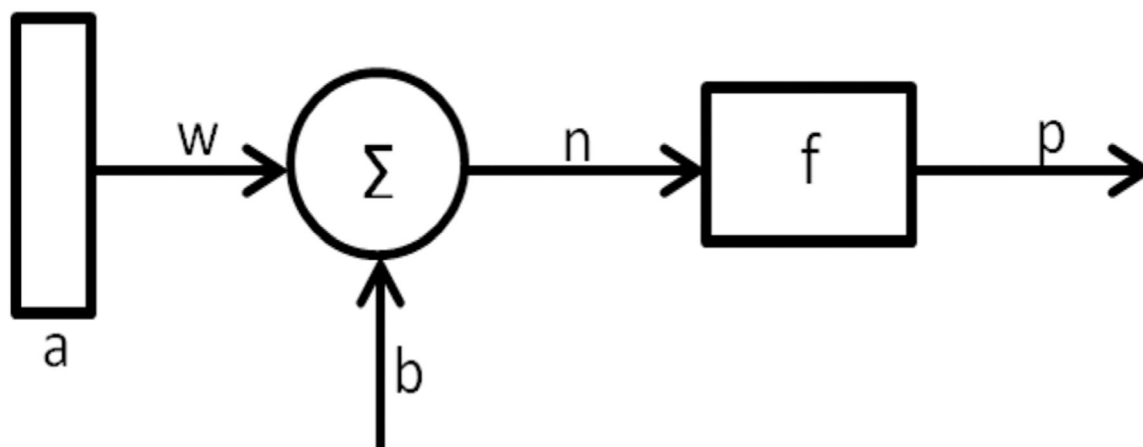


Fig. 2.

A one-input neuron system, where f is the activation function, and a , p , w , and b are the entrance data, outcome, weight, and neuron bias, respectively

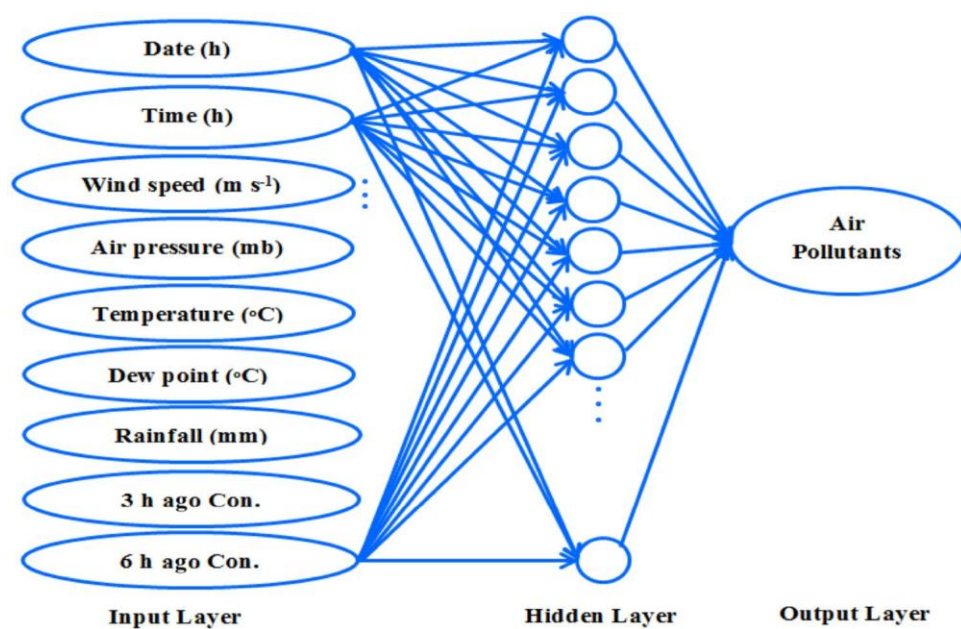


Fig. 3. Structural diagram of ANN with nine inputs, one hidden layer (30 neurons), and one output. “3 h ago Con.” and “6 h ago Con.” refer to concentrations of a particular pollutant 3 and 5 h in advance of the time of the predicted concentration, respectively

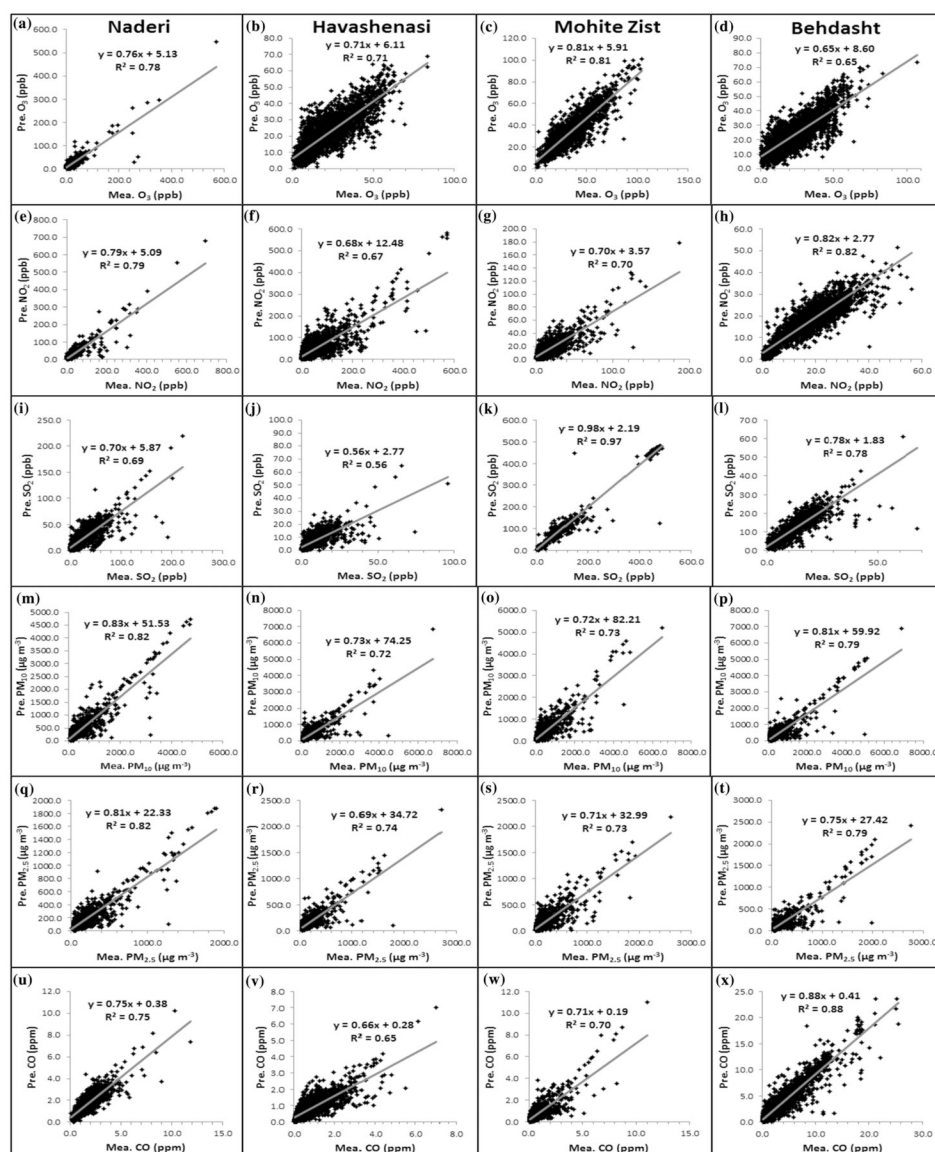


Fig. 4.

Scatterplots showing measured versus predicted concentrations for various pollutants in four stations based on 5% of data used for validation and testing. Data shown for panels in each column refer to the station shown at the top of that column

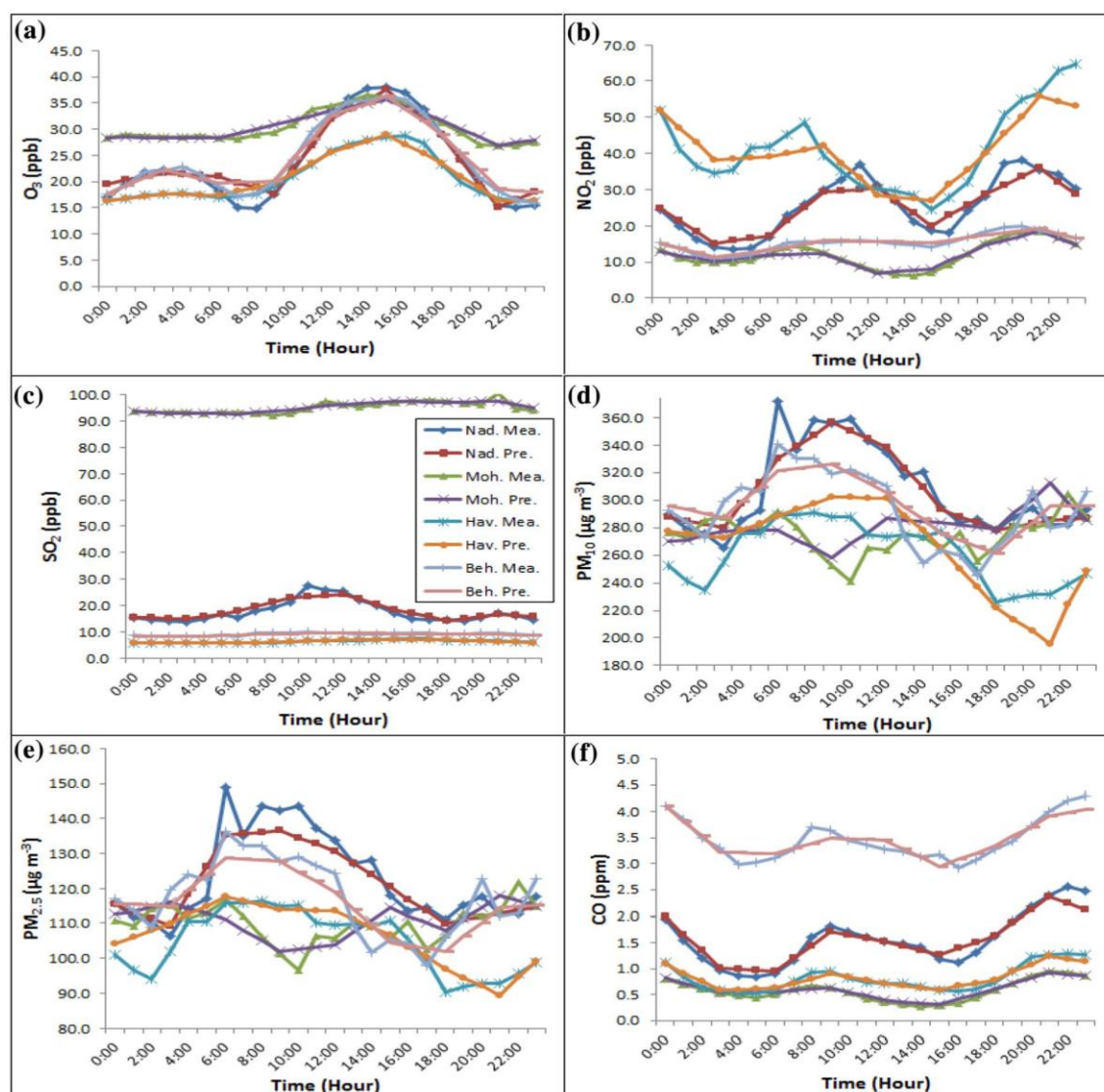


Fig. 5.

Measured and predicted diurnal average concentration of O_3 (a), NO_2 (b), SO_2 (c), PM_{10} (d), $PM_{2.5}$ (e), and CO (f) for the four air quality monitoring stations

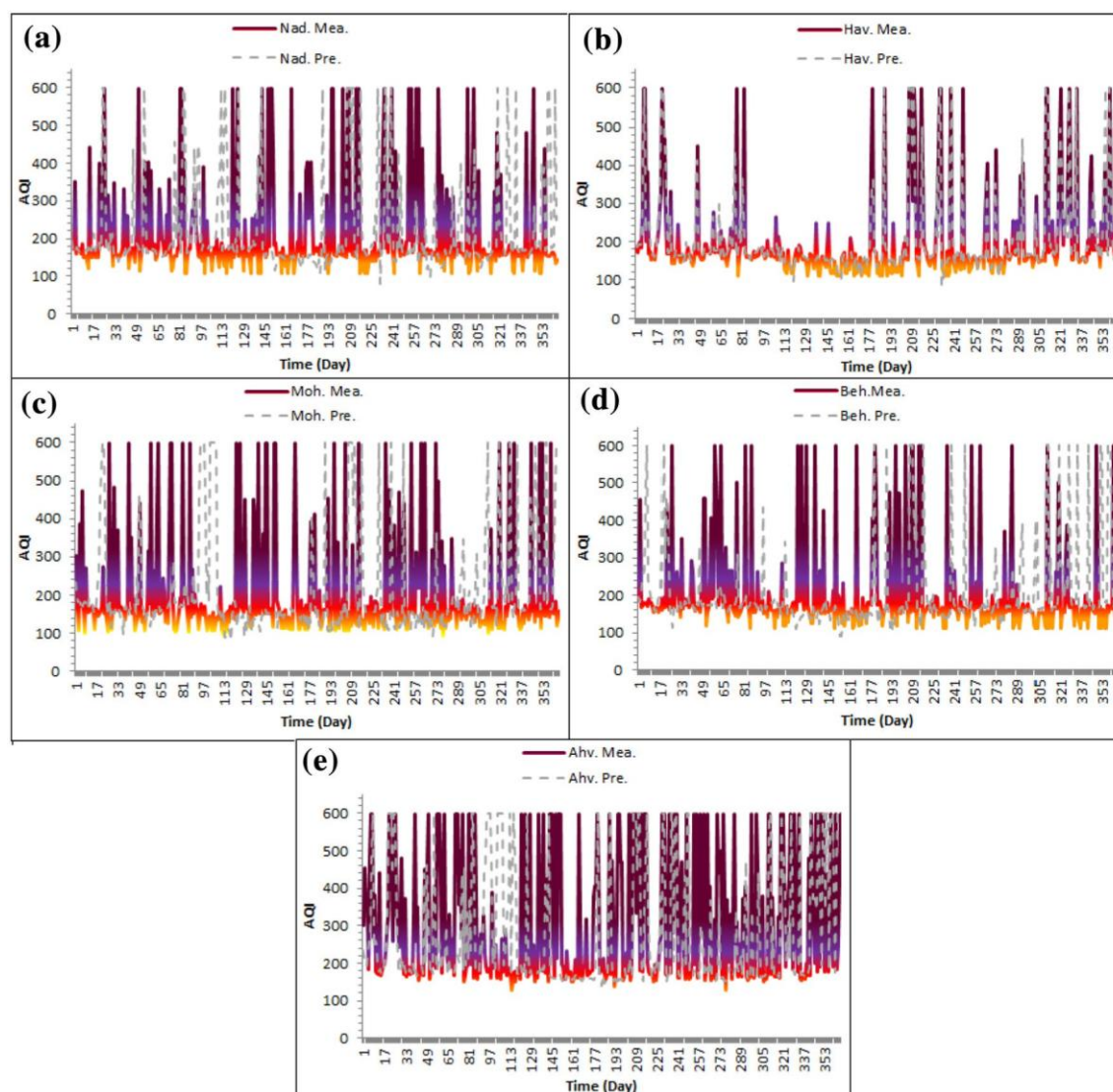


Fig. 6.

Time series of predicted and measured AQI for four air quality monitoring stations in Ahvaz, including all of them averaged, which is representative of the entire city of Ahvaz. Results are shown for the period between August 2009 and August 2010. Values are classified as good (0–50, green), moderate (51–100, yellow), unhealthy for susceptible groups (101–150, orange), unhealthy (151–200, red), very unhealthy (201–300, purple), and hazardous (> 301, Maroon)

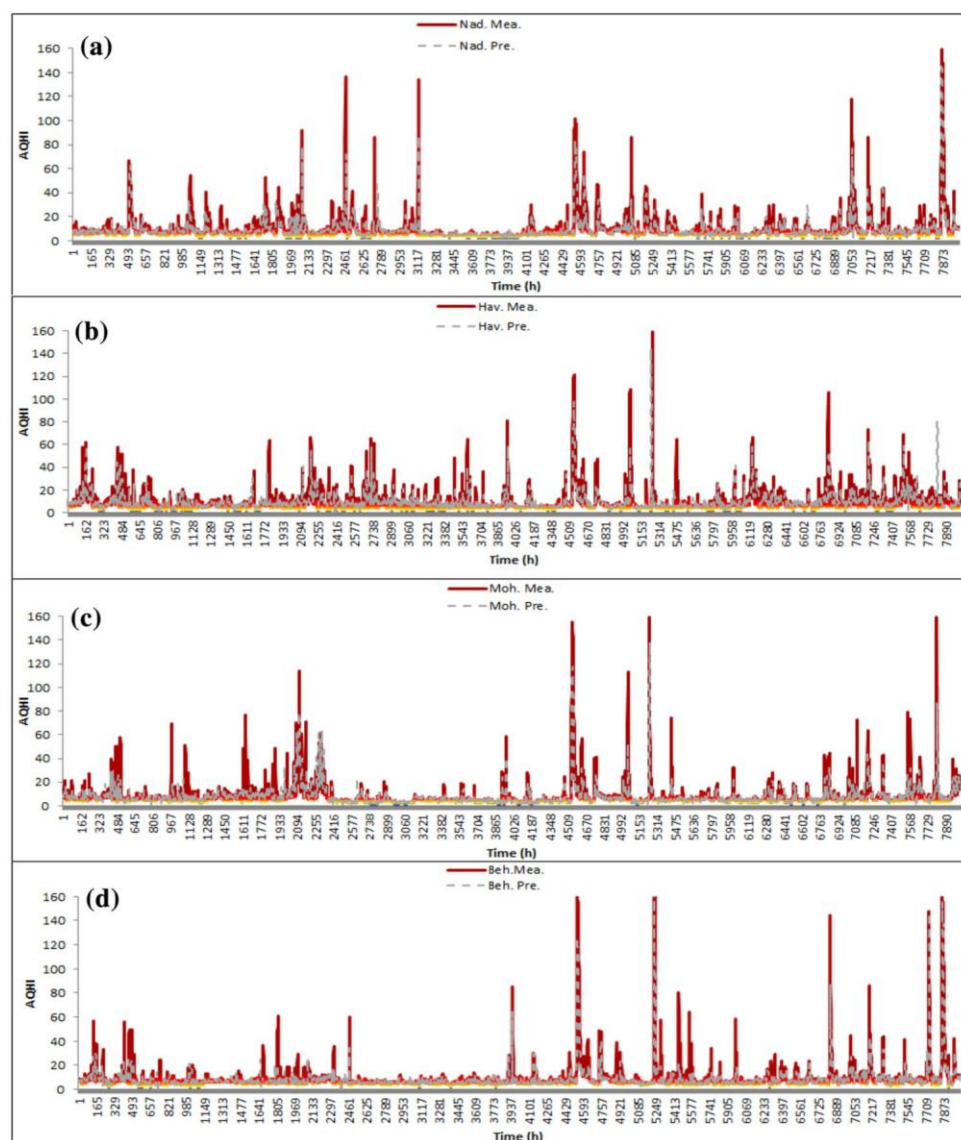


Fig. 7.
Predicted and measured AQHI variations for four air quality monitoring stations from August 2009 through August 2010 in Ahvaz

Table 1

Correlation coefficient (R) and root-mean square error (RMSE) when using 5% and 10% of data for validation and testing of criteria air pollutant concentrations for the four air quality monitoring stations in Ahvaz

Stations	O ₃ (ppb)		NO ₂ (ppb)		SO ₂ (ppb)		PM ₁₀ (µg m ⁻³)		PM _{2.5} (µg m ⁻³)		CO (ppm)	
	R	RMSE	R	RMSE	R	RMSE	R	RMSE	R	RMSE	R	RMSE
<i>5%</i>												
Naderi	0.88	2.8	0.89	4.0	0.83	12.3	0.91	236.6	0.91	80.3	0.87	0.6
Havashenasi	0.84	2.9	0.82	40.9	0.75	5.0	0.85	200.9	0.86	57.5	0.81	0.3
Mohite Zist	0.90	2.8	0.84	13.4	0.99	12.1	0.86	343.3	0.86	123.5	0.84	1.3
Behdasht	0.80	3.0	0.91	4.8	0.88	4.6	0.89	213.4	0.89	69.3	0.94	1.6
Average	0.86	2.9	0.87	15.8	0.86	8.5	0.88	248.5	0.88	82.6	0.87	1.0
<i>10%</i>												
Naderi	0.82	8.2	0.85	19.4	0.82	11.9	0.87	298.3	0.87	80.6	0.85	0.5
Havashenasi	0.83	9.4	0.79	38.1	0.69	5.4	0.84	251.3	0.82	88.2	0.80	0.4
Mohite Zist	0.89	8.4	0.79	11.5	0.99	12.2	0.84	264.6	0.83	136.3	0.81	1.4
Behdasht	0.77	10.3	0.91	4.4	0.87	3.0	0.86	274.0	0.86	90.4	0.93	1.4
Average	0.77	9.1	0.80	18.3	0.83	8.1	0.85	272.0	0.84	98.9	0.74	1.0

Table 2

Range, mean, and standard deviation of input meteorological parameters and measured and predicted criteria air pollutant concentrations (*WS* wind speed, *T* temperature, *T_d* dew point, *P* air pressure, *R* rainfall)

Variable	Unit	Range	Mean	SD
<i>Measured</i>				
WS	m s ⁻¹	[0.0–6.7]	1.1	1.0
T	°C	[3.0–50.2]	27.1	10.1
T _d	°C	[–9.9 to 29.6]	9.8	5.4
P	hPa	[990.7–1025.2]	1009.3	8.1
R	mm	[0.0–34.0]	245.3 ^a	1.6
O ₃	ppb	[0.2–567.8]	25.1	17.4
NO ₂	ppb	[0.1–692.7]	23.5	36.3
SO ₂	ppb	[0.0–488.3]	32.1	54.2
PM ₁₀	µg m ⁻³	[8.0–6900.0]	284.3	421.3
PM _{2.5}	µg m ⁻³	[3.2–2760.0]	113.7	165.5
CO	ppm	[0.0–25.5]	1.6	2.2
<i>Predicted</i>				
O ₃	ppb	[–8.3 to 547.0]	25.0	15.1
NO ₂	ppb	[–156.0 to 681.4]	23.2	31.1
SO ₂	ppb	[–11.3 to 480.2]	32.2	53.7
PM ₁₀	µg m ⁻³	[–492.8 to 6889.6]	287.4	372.3
PM _{2.5}	µg m ⁻³	[–242.3 to 2422.4]	113.6	142.2
CO	ppm	[–1.3 to 23.6]	1.6	2.1

^aThis shows the accumulative volume of rainfall

Number (N) and percentage (%) of days when the measured AQI category does not match the predicted one (moderate (M), unhealthy for sensitive groups (UNSG), unhealthy (UN), very unhealthy (VUN) and hazardous (H^1 and H^2)), in addition to correlation coefficient (R^2) values for the relationship between measured and predicted AQI, and the days out of range (DOR) for measured AQI in Ahvaz

Stations	R^2	M		UNSG		UH		VUH		H^1		H^2		Sum		DOR	
		N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%
Naderi	0.003	-	-	55	15.1	73	20	16	4.4	23	6.3	14	3.8	171	49.6	28	7.7
Havashenasi	0.73	-	-	70	19.2	24	6.6	29	7.9	9	2.5	7	1.9	139	38.1	23	6.3
Mohite Zist	0.004	5	1.4	100	27.4	78	21.4	12	3.3	18	4.9	11	3	224	61.3	31	8.5
Behdasht	0.001	-	-	67	18.4	67	18.4	23	6.3	7	1.9	13	3.6	177	48.5	25	6.8
Ahvaz	0.07	-	-	3	0.8	54	14.8	33	9	36	9.9	30	8.2	156	42.7	71	.5

Number (N) and percentage (%) of hours when the measured AQHI category does not match the predicted one (low (L), moderate (M), high (H) and very high (VH)), correlation coefficient (R^2) value of measured and predicted AQHI and the hours greater and equal than 20 (HGE_{20})

Stations	R^2	L		M		H		VH		Sum		HGE_{20}	
		N	%	N	%	N	%	N	%	N	%	N	%
Naderi	0.81	112	1.4	1239	15.4	913	11.4	602	7.5	2866	35.7	553	6.9
Havashenasi	0.38	368	4.6	1689	21.0	931	11.6	677	8.4	3665	45.6	677	8.4
Mohite Zist	0.70	713	8.9	1336	16.6	852	10.6	381	4.7	3282	40.9	450	5.6
Behdasht	0.78	292	3.6	1417	17.6	864	10.8	466	5.8	3039	37.8	382	4.8